EPSR-test-ex1 Exponential repulsive cores with coordination feedback

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the optional exponential repulsive core that EPSR [1,2] can add to the reference potential of a pair (`rminex` `pcoefex` `coordex` on the first line of a `.pcof` pair entry), and the feedback that adjusts its amplitude `pcoefex` between runs so that the number of close contacts approaches the requested `coordex`. Users apply it to suppress unphysical overlaps that the data alone do not prevent.

Code exercised

Fortran (original, fortran_src/):

- `read_and_save_files.f`: `read_pcof` (reads `rminex` `pcoefex` `coordex`), `save_files_others` (writes them back)
- `epsr_standalone_rev3.f`: `generate_rp` (adds the core to the reference potential), `repulsivept` (`pcoefex` feedback)
- `ato_manipulation_rev4.f`: `atogen` (close-contact count `potex`), `fmolec1`; `molfit.f`: `en_change`

Rust (port, rust/epsr-core/src/):

- `pcof.rs`: `read_pcof`, `write_pcof_pairs`
- `potential.rs`: `generate_rp`, `repulsivept`
- `structure.rs`: `atogen`; `energy.rs`: `fmolec1`, `en_change`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [3] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [4]. Here three pairs get an extra term pcoefex·exp((rminex - r)/(expecf·sqrt(rminex))) (`expecf` 0.3): S-S (rminex 4.0 Å, pcoefex 0, coordex 0.1), water O-O (2.6 Å, 0.5, 0) and water O-H (1.5 Å, 0.2, 0.05). `atogen` counts the average number of intermolecular neighbours closer than `rminex` and `repulsivept` steps `pcoefex` up or down towards the requested coordination number.

Inputs

Copy of EPSR-test with only the `.pcof` changed: the three pair lines `S S`, `O1 O1` and `O1 H1` carry the `rminex` `pcoefex` `coordex` values above (all other pairs keep 0 0 0).

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
Three chained passes, so the updated `pcoefex` values are read back and used. The feedback is active: in the stored Fortran reference after pass 3, `pcoefex` is -4.68E-03 for S-S (from 0), 0.5115 for O-O (from 0.5) and 0.1914 for O-H (from 0.2); the Rust port writes the same values. All 27 output files (`.ato` with random-number state, `.pcof`, `.inp`, `.erg`, g(r), S(Q), potentials, `.out`, `delfit.dat`) are identical after every pass.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[4] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[5] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.